WELCOME

To

REAL ESTATE PRICE PREDICTION

(Machine Learning and Data Science Project)

By our

CSE Dept.
|||rd Year

Team - Techies

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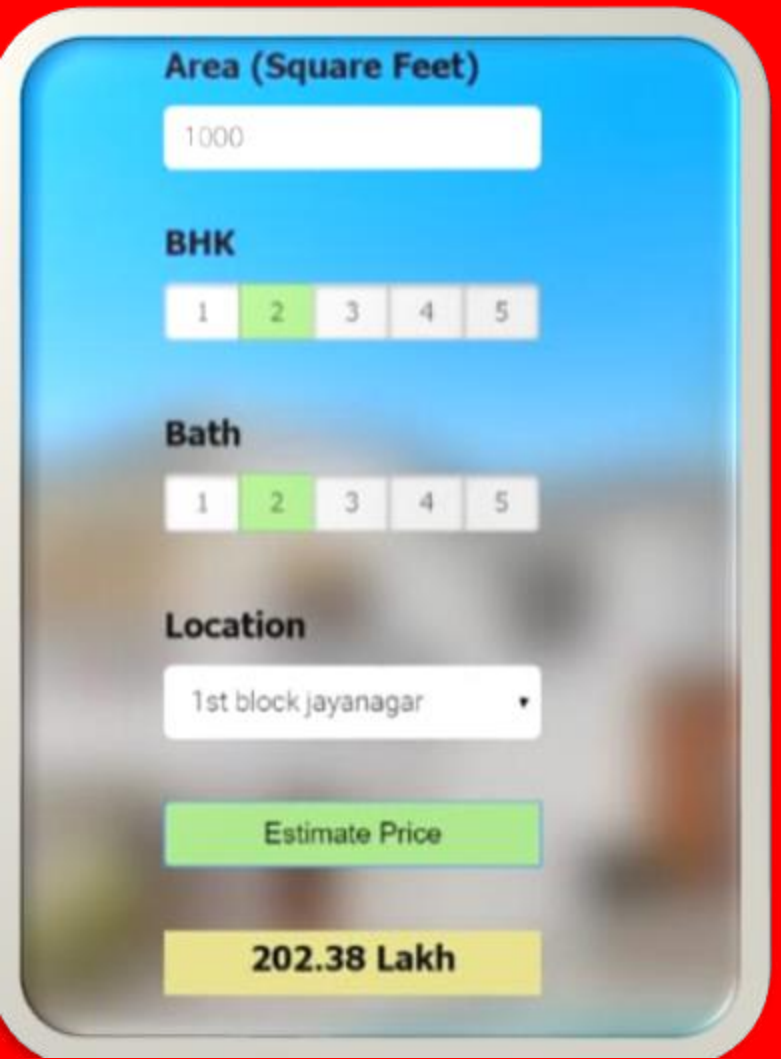
Real Estate Price Prediction Website



Developing a website for real estate price prediction involves multiple stages, from conceptualizing the idea to building the predictive model and designing a user-friendly interface. Here’s a step-by-step guide to help you through the process:

1. Research and Planning

- Understand the market: Study the real estate industry and what influences property prices (e.g., location, size, number of rooms, nearby amenities, etc.).
- Define the goal: Identify the purpose of your website. Is it for general users, real estate agents, or investors?
- Competitor Analysis: Look at similar websites to understand what features they offer, like Zillow or Redfin.

2. Data Collection

- Data Sources: Gather real estate data from reliable sources like government property databases, MLS (Multiple Listing Services), or third-party APIs like Zillow API, Redfin API, or Kaggle datasets.
- Data Features: Collect features such as property location (latitude, longitude), square footage, number of rooms, age, nearby facilities (schools, hospitals), and previous sales history.

3. Data Cleaning and Pre -processing

- Handle missing data: Impute missing values or remove incomplete entries.
- Standardize values: Ensure consistent units (e.g., square feet, square meters).
- Remove outliers: Clean any data points that might disproportionately affect the prediction model.
- Feature Engineering: Create new variables like price per square foot or proximity to amenities, which could improve predictions.

4. Developing the Price Prediction model

- Data Splitting: Split your dataset into training and test sets (usually 80% training, 20% test).
- Model Selection:Linear Regression (if relationships are linear)Decision Trees/Random Forest (for handling non-linear data)
- Model Training: Train the chosen model on your dataset.
- Model Evaluation: Use metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 to assess performance.

5. Back-end Development

- Web Framework: Choose a back-end framework to build your web application:
- Python (Flask/Django): Common for integrating machine Learning
- Database Integration:Use a relational database like PostgreSQL or MySQL for structured data (e.g., property listings).
- For large-scale data, NoSQL databases like MongoDB could also be an option.
- API Development: Create RESTful APIs to handle user requests, such as:
- Uploading property details (size, location)Fetching predicted prices based on input

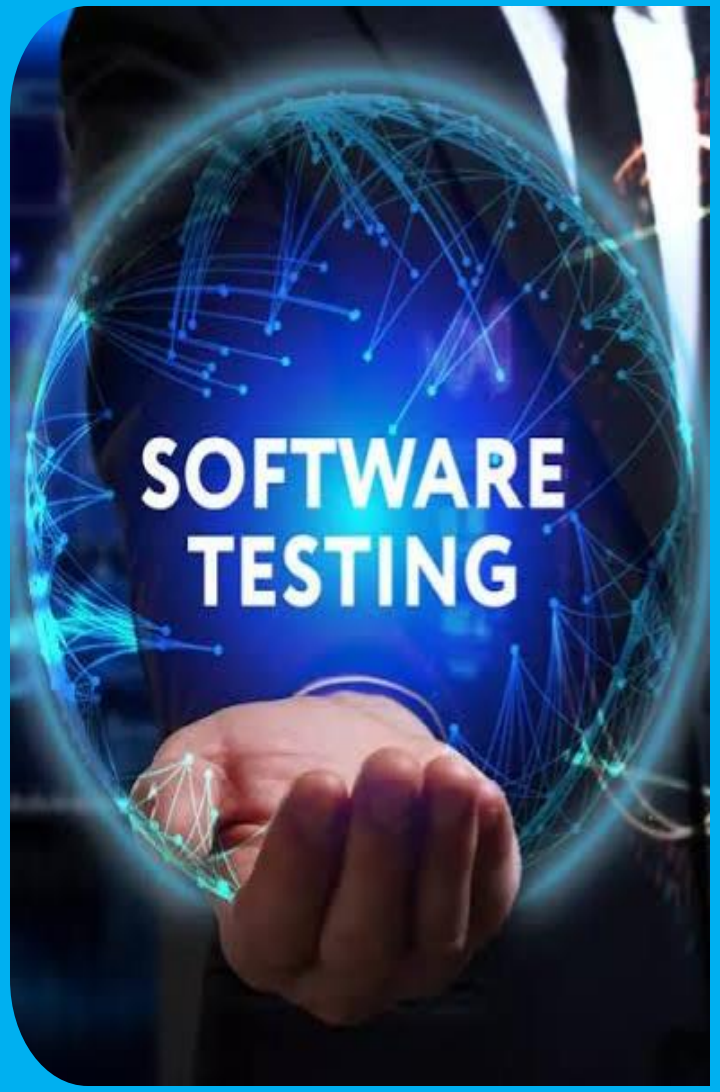
6. Front-end Development

- UI/UX Design: Ensure the website is user-friendly and intuitive:Use wireframing tools like Figma or Sketch to design the interface.Keep forms simple for users to input property data.Provide clear call-to-actions (e.g., “Get Price Estimate”).
- Front-End Framework:Use HTML/CSS/JavaScript for basic design.Use React or Vue.js for building more dynamic and responsive web pages.Use data visualization tools like Chart.js or D3.js to display price trends.
- Responsiveness: Ensure the website is mobile-friendly and works well on all devices.

7. Integration of Machine Learning model

- Deploy Model as API: Use Flask/Django to deploy your model, so the front end can interact with the model.
- Prediction Endpoint: Create an API endpoint that the front end will call to get price predictions based on input data.
- Model Hosting: Host the trained model and API on platforms like AWS, Heroku, or Google Cloud.

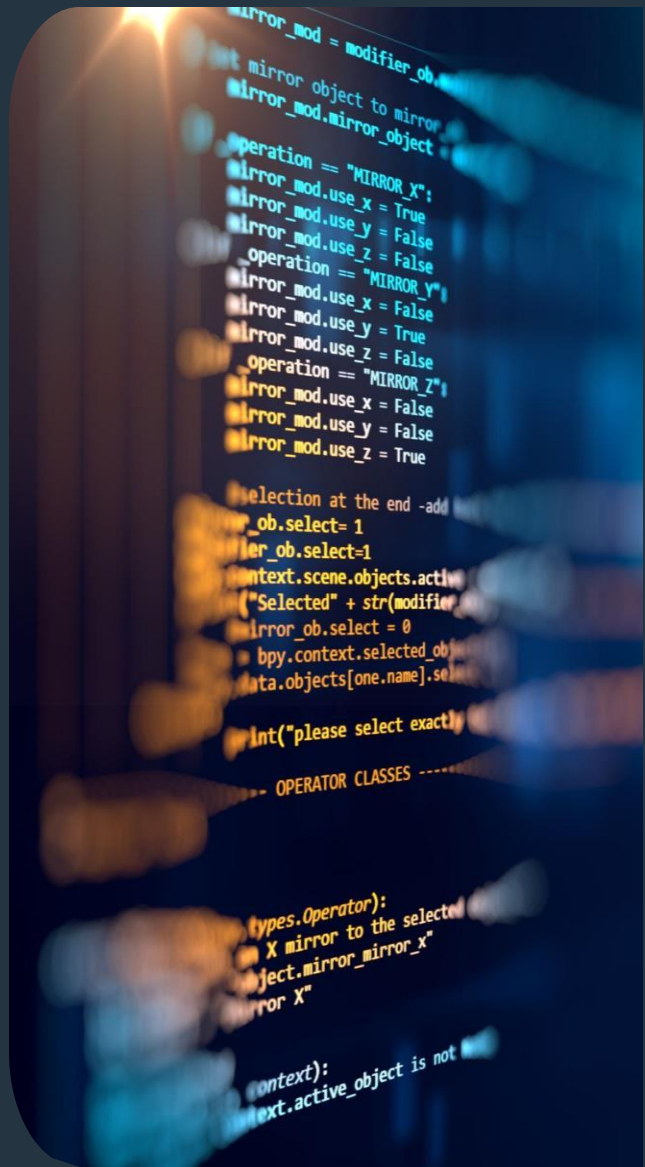
8. Testing and Validation



- Unit Testing: Ensure each part of your code functions as expected (front-end, back-end, and machine learning model).
- Integration Testing: Check that all the components (front-end, back-end, and database) work well together.
- User Testing: Have potential users (e.g., real estate agents) test the website and provide feedback on usability and prediction accuracy.

9. Deployment and Hosting

- Hosting Platforms: Use a web hosting platform like AWS, Heroku, or Google Cloud to host your website.
- Domain Name: Purchase a domain name for your website (e.g., from GoDaddy or Namecheap).
- SSL Certificate: Secure your website with an SSL certificate to ensure data encryption.
- CI/CD Pipeline: Set up a Continuous Integration/Continuous Deployment pipeline using tools like GitHub Actions, Travis CI, or Jenkins to automate testing and deployment.



10. Monitor and Maintenance

- Analytics: Use tools like Google Analytics to track user behavior and interaction with your site.
- Error Tracking: Use error tracking services like Sentry or LogRocket to catch and fix bugs in real time.
- Model Retraining: Periodically retrain the model as new real estate data becomes available to keep the predictions accurate.

Tools and Technologies

Technology Stack Summary:

- Front-end: HTML, CSS, JavaScript(React/Vue.js)
- Back-end: Python (Flask/Django), Node.js
- Database: PostgreSQL/MySQL, MongoDB
- Modeling: Scikit-learn, TensorFlow, XGBoostHosting: AWS, Google Cloud, Heroku

By following these steps, you will be able to create a functional website that predicts real estate prices based on user input.

Team - Techies

CSE department,3rd year

Thank you